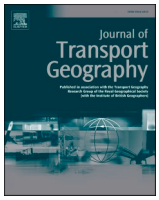
Journal of Transport Geography 128 (2025) 104387



Contents lists available at ScienceDirect

Journal of Transport Geography

journal homepage: www.elsevier.com/locate/jtrangeo



Navigating spatial inequities in freight truck crash severity via counterfactual inference in Los Angeles

Yichen Wang [a], Hao Yin [a], Yifan Yang [b], Chenyang Zhao [c], Siqin Wang [d,e,f,*]

[a] The Research Center for Digital Production and Smart Logistics, Tsinghua University, Department of Logistics and Transportation at Tsinghua University, China
[b] Geospatial Exploration and Resolution (GEAR) Lab, Department of Geography, College of Arts & Sciences, Texas A&M University, USA
[c] Artificial General Intelligence Lab, University of California, Los Angeles, Computer Science department at University of California, Los Angeles, USA
[d] Spatial Sciences Institute, University of Southern California, USA
[e] School of Science, RMIT University, Melbourne, Victoria, Australia
[f] School of the Environment, University of Queensland, Brisbane, Australia

ARTICLE INFO

Keywords:
Geographical equity
Spatial justice
Freight truck-related crash
Deep counterfactual inference
Policy intervention

ABSTRACT

Freight truck-related crashes pose significant challenges, leading to substantial economic losses, injuries, and fatalities, with pronounced spatial disparities across different regions. This study adopts a transport geography perspective to examine spatial justice concerns by employing deep counterfactual inference models to analyze how socioeconomic disparities, road infrastructure, and environmental conditions influence the geographical distribution and severity of freight truck crashes. By integrating road network datasets, socioeconomic attributes, and crash records from the Los Angeles metropolitan area, this research provides a nuanced spatial analysis of how different communities are disproportionately impacted. The results reveal significant spatial disparities in crash severity across areas with varying population densities, income levels, and minority populations, highlighting the pivotal role of infrastructural and environmental improvements in mitigating these disparities. The findings offer insights into targeted, location-specific policy interventions, suggesting enhancements in road infrastructure, lighting, and traffic control systems, particularly in low-income and minority-concentrated areas. This research contributes to the literature on transport geography and spatial equity by providing data-driven insights into effective measures for reducing spatial injustices associated with freight truck-related crashes.

1. Introduction

The demand for road freight transportation has increased significantly in the past few decades due to globalization and the rise of e-commerce, leading to a higher concentration of freight movement in specific geographical areas and corridors (Jain et al., 2020; H. Wang et al., 2021; Yu et al., 2025). The spatial distribution of freight activities has created varying exposure levels to freight truck-involved crashes across different communities, increasing the likelihood of such incidents in certain regions. The Federal Motor Carrier Safety Administration (FMCSA) noted that from 2009 to 2019, the number of large trucks involved in fatal crashes increased by 47 %, reflecting not only the increased presence of trucks on the road but also the geographical disparities in associated risks (FMCSA, 2021). Crashes involving freight trucks are more severe than those involving smaller vehicles. Due to the size and weight, freight trucks inflict more extensive damage, resulting in higher fatality and serious injury rates. In 2020, large trucks were involved in 9 % of all fatal crashes, while they only account for 4 % of all registered vehicles and 7 % of total vehicle miles traveled, according to the National Highway Traffic Safety Administration report (NHTSA, 2021). These statistics underscore the necessity of understanding the spatial patterns of freight truck-involved crashes and their implications for different regions and communities. Freight truck-involved crashes can cause significant economic losses due to delays, damage to goods, and increased transportation costs (Fountas et al., 2020; Hossain et al., 2023), effects that may disproportionately impact certain geographic areas.

Given the rising severity of freight truck-involved crashes, numerous

* Corresponding author at: Spatial Sciences Institute, Dornsife College of Letters, Arts and Sciences, University of Southern California, 3616 Trousdale Parkway, AHF B57C, Los Angeles, CA 90089-0374, USA.
E-mail addresses: wang-yc22@mails.tsinghua.edu.cn (Y. Wang), yinh24@mails.tsinghua.edu.cn (H. Yin), yyang295@tamu.edu (Y. Yang), zhaochenyang@cs.ucla.edu (C. Zhao), siqinwan@usc.edu (S. Wang).

https://doi.org/10.1016/j.jtrangeo.2025.104387
Received 11 November 2024; Received in revised form 15 June 2025; Accepted 11 August 2025
Available online 20 August 2025
0966-6923/© 2025 Published by Elsevier Ltd.

studies have examined factors influencing the temporal and spatial distribution of these accidents, as well as the severity of injuries sustained based on the different weather conditions and road types (Ahmed et al., 2018; Yan et al., 2024; Yang et al., 2021b). By examining the spatiotemporal distributions of freight truck-involved crashes, a series of prevention strategies have been proposed and implemented based on the identified risk factors to enhance overall road safety, including management strategies such as zone-based restrictions during peak hours (Sze and Song, 2019) but also from the traffic design (Bassan, 2016) and urban planning (Dumbaugh and Rae, 2009; Yan et al., 2024) perspectives. However, these strategies often overlook the spatial equity implications, where certain communities, particularly low-income or minority populations (Najaf et al., 2018), may bear a disproportionate burden of freight-related risks due to their geographical location near major freight corridors or industrial zones. For example, Park and Park (2022) found that snowy and wet road surfaces, construction zones, and pavement defects significantly increase crash risk when assessed with multilevel mixed-effects models. Yan et al. (2024) examined strategies to improve freight truck safety on urban non-highway roads and areas adjacent to highway access. Despite these contributions, there remains a gap in understanding how geographical disparities and spatial inequities influence the distribution and severity of freight truck-involved crashes (Faisal Habib et al., 2024; Yang et al., 2021a). The analysis and prevention of freight truck crashes require empirical research based on solid evidence (McDonald et al., 2019). Consequently, most existing studies have merely identified associations between factors and crash locations, without quantifying the magnitude or interplay of these factors across different geographical contexts, leaving a gap between a comprehensive understanding of these relationships and the development of effective, location-specific mitigation strategies.

The impact mechanism across geospatial contexts has been inadequately explored in past research. As the specific contributing factors to freight truck–related crashes can differ substantially by region, owing to variations in road network configurations, socioeconomic characteristics, and environmental conditions, their effects may not be uniform across communities and geographic settings. For example, areas with higher population densities or lower-income communities may face different risk factors compared to more affluent suburban regions or industrial zones (Yu et al., 2024). Therefore, exploring heterogeneity is crucial to understanding spatial justice in freight truck-involved crashes. It provides a lens through which we can identify which communities are more vulnerable due to the spatial distribution of risk factors and the intersection of environmental, economic, and infrastructural disparities.

While recent studies have put increasing emphasis and begun to explore the influencing mechanism of traffic crashes (Elvik, 2024; Shinar and Hauer, 2024; Ulak and Ozguven, 2024), studies specifically focusing on freight truck accidents from a transport geography perspective are scarce. This lack of targeted research makes it challenging to understand causative factors unique to freight truck-related crashes, which are often more complex due to the trucks' route choice behaviors and complicated interactions with built environments such as road network patterns and land use patterns. Moreover, these studies have not adequately addressed the spatial equity concerns and counterfactual scenarios, which ask whether a crash would still have occurred if certain conditions had been different in specific geographical contexts. Without this understanding, it is difficult to evaluate the potential impact of proposed interventions or changes in conditions on different regions or communities. Counterfactual analysis is critical for designing effective prevention strategies that can adapt to real-world spatial dynamics.

To address the research gap in understanding the factors influencing freight truck-involved crashes and their spatial distribution, we propose a deep counterfactual inference model from a transport geography standpoint, offering a more precise analysis of relationships and the potential effects of different interventions on crash severity across diverse geographic areas. This model allows for the inference of how changes in road conditions, driver behaviors, and environmental factors could affect crash outcomes in various spatial contexts. By simulating counterfactual scenarios, our model enables a detailed comparison between the observed crash data and hypothetical interventions, providing insight into the effectiveness of specific safety measures targeted at specific regions. The study leverages empirical data from the Los Angeles Metropolitan area to evaluate cross-sectoral strategies, focusing on interventions such as improved traffic management systems and road infrastructure enhancements in areas identified as high-risk zones due to spatial inequity factors. This approach enables us to quantify the effectiveness of interventions like lighting and road control device optimization in reducing the severity of freight truck crashes and addressing geographical disparities.

This research contributes to both theoretical and practical domains. Theoretically, we introduce a deep counterfactual inference model that expands the analytical capability for investigating factors underlying freight truck crashes within a transport geography and spatial equity framework. It represents a significant advancement in traffic safety research by providing a robust structure for evaluating how changes in various conditions influence crash severity across different locales. Practically, by simulating the outcomes of diverse policy interventions, the model enables the development of targeted strategies to reduce crash severity and mitigate spatial inequity. This approach clarifies how adjustments in road infrastructure and driver behaviors can alleviate freight truck risks, particularly in communities disproportionately affected, thereby improving road safety through a spatial equity lens.

The remainder of this study is organized as follows. Section 2 reviews related studies, and Section 3 presents the entire methodology framework. In Section 4, we introduce the data sources, study area, and model configuration. Section 5 illustrates the analyzed result, and Section 6 discusses the related policies for strengthening spatial equity. Section 7 concludes the entire study.

2. Literature review

2.1. What factors determine the spatial distribution of freight truck-involved crashes?

Multiple interrelated factors shape the spatial distribution of freight truck-involved crashes, which can be grouped into four primary categories according to the existing literature: road infrastructure, traffic status, weather conditions, and socioeconomic and demographic factors (Hong et al., 2019; Taylor et al., 2018; Yan et al., 2024). Road infrastructure plays a critical role in determining where freight truck crashes are likely to occur (Haddad et al., 2023; Ramiani and Shirazian, 2020). The design and quality of infrastructure can either mitigate or exacerbate crash risk. Research has shown that road design and patterns significantly influence crash rates. For instance, roads with complex intersections, sharp curves, and inadequate lane widths create hazardous conditions for freight trucks, which require more space and time to drive (Bassan, 2016). Poorly designed roadways increase the likelihood of crashes by limiting truck drivers' ability to respond to sudden changes in traffic conditions. Guo et al. (2017) were among the first to report on road network pattern factors in pedestrian-related crashes. Yan et al. (2024) found that the regional road network's centrality index accounts for at least 17.11 % of the variation in the spatial distribution of freight truck-involved crashes in the Los Angeles metropolitan area. The presence of dedicated freight lanes is another crucial factor in this category. These lanes help reduce interactions between trucks and smaller vehicles and pedestrians, thereby minimizing potential conflicts and enhancing overall safety. Segregated lanes for trucks lead to fewer collisions by allowing freight vehicles to maintain consistent speeds and reducing the need for frequent lane changes (Shin, 2024a; Jianyu Wang et al., 2020). These findings emphasize that investing in road design, maintenance, and dedicated freight infrastructure can significantly reduce crash frequency, underscoring the importance of infrastructure

planning in traffic safety strategies (Yuan and Wang, 2021). However, these studies also underscore several limitations and challenges. While dedicated lanes and improved infrastructure can enhance safety, the implementation of such measures often faces practical constraints, including limited urban space and high construction costs. Furthermore, the effectiveness of these strategies is influenced by other factors such as driver behavior (Mehdizadeh et al., 2019; Newnam et al., 2018; Rashmi and Marisamynathan, 2023), which cannot be fully addressed through road infrastructure.

Traffic conditions, particularly volume and congestion, are pivotal in influencing the spatial distribution of freight truck crashes. High traffic volumes often correlate with increased crash risks, as they can lead to congestion (Rahimi et al., 2019), which exacerbates the likelihood of accidents. C. Wang et al. (2022a) demonstrated that congested conditions increase not only crash frequency but also crash severity. In heavy traffic, trucks have limited maneuverability, making safe navigation and collision avoidance more difficult. Speed variability is another critical factor (Madarshahian et al., 2024). In regions with mixed traffic flows and inconsistent speed limits, the risk of crashes increases (Duvvuri et al., 2022). Choudhary et al. (2018) have shown that regions with frequent speed changes and stop-and-go traffic tend to experience higher rates of truck-involved crashes. Such variability creates hazardous situations in which trucks struggle to maintain safe distances, leading to collisions (Al-Bdairi and Hernandez, 2020; Song and Fan, 2021).

Weather conditions are a well-documented factor affecting the spatial distribution of freight truck crashes (Becker et al., 2022; Kim et al., 2024; Li et al., 2020; Naik et al., 2016). An 18-year longitudinal analysis from 1994 to 2012 in the whole U.S. has concluded that adverse weather, such as rain, snow, and fog, reduces visibility and road traction, significantly increasing crash likelihood (Saha et al., 2016). Park and Park (2022) employed multilevel mixed-effects models to demonstrate that adverse weather conditions contribute substantially to the occurrence of freight truck crashes. These conditions can lead to slippery roads and impaired driver visibility, making it more challenging for truck drivers to control their vehicles and react to sudden changes in traffic conditions. Seasonal variability also plays a vital role, with winter months often experiencing higher crash rates due to snow and ice (Okafor et al., 2022; Wei et al., 2022).

The impact of weather on freight truck crash distribution underscores the need for weather-responsive traffic management strategies and prevention infrastructure design that considers seasonal changes to improve safety. The interaction between traffic status and weather conditions further complicates this dynamic (Okafor et al., 2022). Adverse weather conditions impact traffic flow, leading to congestion and reduced visibility, which increases crash risks for freight trucks. Therefore, understanding the interplay between these factors is essential for developing comprehensive traffic management strategies that enhance road safety.

The impact of e-commerce on the spatial distribution of freight truck crashes has also become an emerging concern. McDonald et al. (2019) discuss how the shift of freight vehicles onto local streets and arterials due to e-commerce has led to rising crash rates in urban areas. Her study found that between 2005 and 2015, urban freight-related fatality rates increased by 17 %, with significant rises observed in crashes not occurring on interstates. Nonfatal injury rates for urban freight-involved crashes rose even more rapidly, increasing by 45 % between 2009 and 2015. This shift is partly attributed to increased freight volumes and changing patterns of freight traffic, as commercial and residential areas face higher levels of congestion and risk. These findings highlight the need for improved infrastructure planning and safety measures in urban freight corridors (Yang et al., 2021b).

Socioeconomic and demographic factors are often correlated with transportation infrastructure quality and traffic conditions, therefore impacting the spatial distribution of freight truck-involved crashes (Yuan et al., 2023; Yuan and Wang, 2021). Areas with high economic activity and dense industrial and logistical land use typically experience more freight traffic, leading to increased crash risks (Yan et al., 2024). Dumbaugh and Rae (2009) explored how urban design and land use patterns impact crash distribution, emphasizing the need for integrated planning that considers both economic and traffic dynamics. Demographic disparities also play a critical role in shaping these patterns. Regions with higher concentrations of minority and low-income populations often face poorer road conditions and higher crash risks. Yan et al. (2024) have highlighted the disparities in road network patterns across different socioeconomic groups, suggesting targeted interventions to address these inequalities. These findings underscore the importance of equitable infrastructure development to improve safety for all road users, particularly in underserved communities. The interplay between the factors underscores the complexity of freight truck crash dynamics and the need for novel approaches to consider this latent interplay.

2.2. Freight truck-involved crash estimation models

Existing research on freight truck-involved crashes has employed various models to estimate and analyze crash occurrences, each capturing the complexity and dynamics of influencing factors, mainly including three primary groups: traditional statistical models, machine learning-based models, and hybrid models (Hossain et al., 2024a). Statistical models have long been the foundation of crash estimation because they allow researchers to identify and quantify relationships between crash occurrences and influencing factors (Yuan and Wang, 2021). Existing studies have adopted statistical models, such as the zero-inflated and negative binomial models, to address the sparsity and overdispersion commonly found in freight truck-involved crash data (Lord et al., 2005; Pew et al., 2020). Yang et al. (2021a) employed a zero-inflated Poisson model to reveal a negative correlation between distance to freight transport centers and the density of fatal or serious injury crashes. The freight truck-involved crashes also presents spatial autocorrelation patterns, hence, Yuan and Wang (2021) applied a spatial lag model (SLM) to capture spatial dependencies between census tracts and provide empirical evidence of spatial inequities in crash occurrences.

While statistical models provide clear and interpretable results, they often rely on assumptions of linearity and independence among factors, which may not fully capture the complex interactions inherent and the nonlinear relationship in crash data. Yan et al. (2024) addressed this limitation by employing Extreme Gradient Boosting models and integrating freight-facility features extracted from high-resolution satellite imagery to mitigate zero-inflation issues. A previous work by them offered another approach, which provided a concept of freight truck crash hazards (FTCHs) to express the latent hazards of the crashes and further designed Bayesian deep learning models to understand road network patterns in shaping local truck crashes' severity (Yuan et al., 2023).

Different from the above-supervised machine learning methods, combined with the interpretable framework, such as Shap theory (Yang et al., 2021a), Hossain et al. (2024b) employed an unsupervised learning algorithm to analyze crash data, revealing distinct patterns and spatial clusters that identify high-risk crash areas at high-speed locations. Furthermore, recent studies have attempted to combine the advantages of traditional statistical models with advances in machine learning-based methods. Hossain et al. (2024a) employ a two-step hybrid modeling approach using Extreme Gradient Boosting to identify key variables and Correlated Random Parameter Ordered Probit with Heterogeneity in Means (CRPOP-HM) to predict crash injury severity, concluding that factors such as collision type, contributing factors, presence of passengers, location, and weekend driving significantly increase the risk of severe crashes involving older drivers on high-speed roads.

While these models provide significant insights into freight truck-

involved crash estimation by capturing complex patterns and identifying key risk factors, they often lack the ability to perform counterfactual inference. The ability to simulate "what-if" scenarios is crucial for identifying the causes of freight truck-involved crashes and designing effective interventions. Meanwhile, although machine learning models excel in handling large datasets and capturing nonlinear relationships, they typically operate as "black boxes," lacking interpretability. This gap in understanding the mechanisms behind crash occurrences limits the ability to predict the effects of potential interventions accurately. Hence, our research aims to address these challenges by integrating counterfactual inference techniques within a deep learning framework. By leveraging advanced modeling approaches, we seek to improve the estimation of freight truck crash risks and uncover the relationships driving these incidents. This will provide more actionable insights for policymakers and urban planners, ultimately contributing to enhanced safety measures and more effective strategies to reduce freight truck-involved crashes.

3. Methodology

3.1. Deep counterfactual inference framework

To address the challenges of understanding and quantifying the effects of various factors on freight truck crashes, we propose a deep counterfactual inference (DCI) framework that simulates hypothetical scenarios (counterfactuals) to estimate how changes in specific factors might influence crash occurrences for prevention. Building upon advances in deep learning, DCI integrates counterfactual reasoning with nonlinear representation learning to overcome the limitations of traditional methods such as propensity score matching or logistic regression in handling high-dimensional, nonlinear relationships and unobserved confounders (Hartford et al., 2017; Johansson et al., 2016; Pawlowski et al., 2020).

The DCI framework estimates how the probability distribution of crash severity would differ under alternative configurations of influencing factors, quantifying the influence of altered conditions or interventions by comparing simulated counterfactuals to factual outcomes. This approach is suited to transportation safety analysis, wherein observational data often suffer from confounding bias and where it is neither ethical nor practical to test interventions such as modifying roadway lighting in real time. Architecturally, DCI employs a multi-task deep learning design consisting of an input component encoding observed covariates, a calibration component(balancing factual and counterfactual representations, and a shared representation component learning latent features common to both tasks, followed by two task-specific heads that predict outcomes under factual and counterfactual scenarios in Fig. 1.

For each freight truck-related crash occurrence i, the outcome variable $\mathbf{Y}_i$ denotes the severity of the crash in an ordered form and covariates $\mathbf{X}_i$ represents the vector representing the observed conditions and influencing factors at the time of the crash (e.g., Lighting condition, weather condition, control device condition). There are two categories in the covariates $\mathbf{Z}_i$, one is the confounders $\mathbf{X}_i$, influencing not only the outcome but also other covariates, the other category is treatment variables $\mathbf{T}_i$. And the counterfactual inference results are defined as $\mathbf{Y}_i^*$ and $\mathbf{T}_i^*$ for the collaboration learning process of the counterfactual scenario in the DCI framework, obtained via methods such as propensity score matching to guide counterfactual learning. In addition, $\mathbf{U}_i$ is utilized to denote unobserved factors influencing freight truck crashes.

The network structure, training process and parameter configuration are presented in Appendices A and B. The trained DCI model enables to estimate the impact of changing any combination of factors $\mathbf{X} \in \mathbf{T}$ on the probabilities of various crash severity levels. Since the outputs are probability distributions over severity levels, we can address two key questions to denote the counterfactual effects: (1) Does changing a given set of factors $\mathbf{X} \in \mathbf{T}$ cause the freight truck-related crash's severity level to change between factual and counterfactual scenarios? (2) If the severity level remains the same between factual and counterfactual scenarios, does the change in factors $\mathbf{X} \in \mathbf{T}$ alter the probabilities associated with that severity level?

For the first question, we define the severity level of individual treatment effect (ITE) $\widehat{ITE}_{i,level}(\mathbf{X})$, which indicates whether the predicted severity level changes when the influencing factors $\mathbf{X}$ are altered. The severity level with the highest predicted probability under factual and counterfactual conditions is selected as the output results. Mathematically, this can be expressed as Eq. (1). The corresponding average treatment effects (ATE) can be denoted as Eq. (2).

$$\widehat{ITE}_{i,level}(\mathbf{X}) = \underset{k\in K}{\operatorname{argmax}}\widehat{y_{i,k}^{CF}} - \underset{k\in K}{\operatorname{argmax}}\widehat{y_{i,k}^{F}} \tag{1}$$

$$\widehat{ATE}_{level} = \frac{1}{N}\sum_{i=1}^{N}\widehat{ITE}_{i,level}(\mathbf{X}) \tag{2}$$

where $\widehat{y_{i,k}^{CF}}, \widehat{y_{i,k}^{F}}$ are the predicted probabilities for severity level k of the factual and counterfactual scenarios, respectively. N denotes the sample number.

For another question, when the severity level does not change ($\widehat{ITE}_{i,level}(\mathbf{X}) = 0$), we introduce the probability difference individual treatment effect $\widehat{ITE}_{i,probability}(\mathbf{X})$ to represent the counterfactual effect. This measures the difference in the predicted probabilities of the current same severity level between the factual and counterfactual scenarios, which can be defined as Eq. (3). The corresponding ATE can be calculated as Eq. (4).

$$\widehat{ITE}_{i,probability}(X) = \underset{k\in K}{max}\widehat{y_{i,k}^{CF}} - \underset{k\in K}{max}\widehat{y_{i,k}^{F}} \tag{3}$$

$$\widehat{ATE}_{probability} = \frac{1}{N}\sum_{i=1}^{N}\widehat{ITE}_{i,probability}(\mathbf{X}) \tag{4}$$

Where K represents the set of the crash severity level. By utilizing these two counterfactual indexes, we can comprehensively assess the counterfactual effects of changing influencing factors on both the severity level and the associated probabilities of freight truck-related crashes.

3.2. Outcome, treatment, and confounder variables

This analysis of freight truck-involved crashes organizes variables into three key categories: outcome, treatment, and confounder variables. These categories form a framework that helps to understand the factors influencing crash severity, with a focus on potential spatial equity concerns and disparities in crash outcomes.

The crash severity level is chosen as the outcome variable, which measures the injuries degree resulting from freight truck crashes off freeways, measuring injury extent from minor to fatal (Taylor et al., 2018). Modeling crash severity helps evaluate intervention effectiveness and understand accident consequences (Shin, 2024a). By identifying and quantifying the factors that cause higher crash severity, policymakers can develop more targeted and effective safety measures, especially in regions with potential socio-economic or road infrastructural spatial equity concerns.

Treatment variables include factors which have a direct effect on

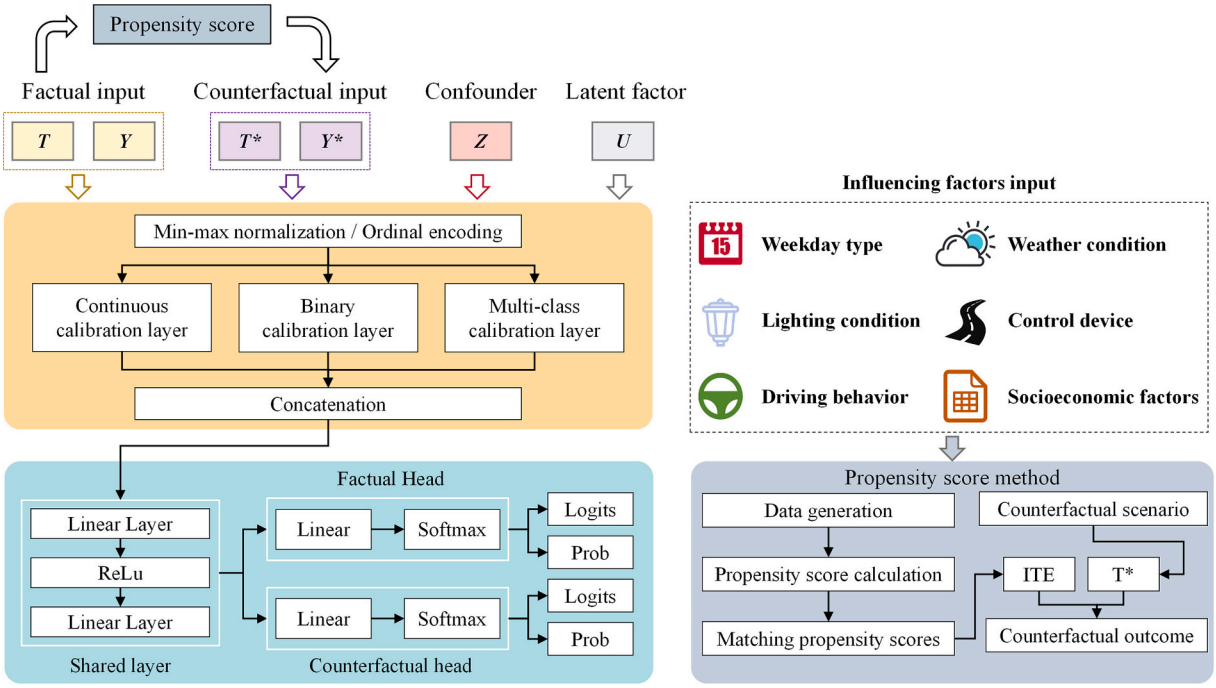


Fig. 1. The architecture of the DCI framework.

crash severity and can inform policy interventions (Das et al., 2022; Hong et al., 2019). These variables encompass road environment factors, such as lighting, weather, and traffic control device conditions, which influence road safety, particularly for freight trucks that require more stopping distance and are harder to maneuver (Uddin and Huynh, 2017). Speed dummy and collision severity[1] are also included, reflecting the presence of speeding violations and the severity of the collision, both of which directly impact crash outcomes (Newnam and Goode, 2015). The inclusion of lane number, bridge dummy, and tunnel dummy variables accounts for the influence of specific road features, such as lane count and special road types, which can affect crash severity (Shin, 2024b). Traffic volume indicator such as the annual average daily traffic volume density (AADT density) help address spatial imbalances in crash severity caused by uneven traffic distributions across regions (Yuan and Wang, 2021).

Road network variables, such as PageRank proportion, degree centrality, cluster coefficient, average road segment length, and intersection density, capture the structural properties of the road network that influence crash risks. These variables provide a more comprehensive view of how network features influence crash outcomes. The involvement of pedestrians, cyclists, and motorcyclists in freight truck crashes presents a heightened risk due to the vulnerability of road users (Jueyu Wang et al., 2022b). Protecting them through dedicated road infrastructure, such as safer crosswalks and designated lanes should inform policy decisions.

The confounder variables include socioeconomic and land use attributes, which impact crash severity levels indirectly. These variables are crucial for identifying spatial equity concerns, as they often reveal disparities in how different regions are impacted by freight truck crashes. Socio-economic attributes such as population density and mean household income provide a view into regional crash severity differences (Yuan and Wang, 2021). Higher population density can increase traffic interactions and the likelihood of severe accidents, while lower-income areas often receive less infrastructure investment and thus face higher crash severity. The model also includes the specific percentages of each racial and ethnic group at the census block group level, allowing for a deeper understanding of how different racial and ethnic groups are disproportionately affected by freight truck–related crashes.

Sector attributes including the proportions of manufacturing, wholesale trade, Retail trade, transportation and warehousing jobs reflect the intensity of freight-related activity in specific regions (Shin, 2024b). Regions with greater concentrations of freight traffic, especially those tied to transportation and warehousing, are at elevated risk of severe crashes due to more heavy-vehicle presence.

4. Study area, data sources, and descriptive analysis

This research employs the Los Angeles Greater Metropolitan Area (LAMA), which spans over 33,000 mile^2 and includes a population of approximately 18.7 million people (Yoon et al., 2018) as the study area and uses ten years of freight truck-related crash data from 2012 to 2022, obtained from the California Statewide Integrated Traffic Records System (SWITRS) database (the Southern California Association of Governments, 2022).[2] This study focuses on crashes occurring on non-freeway roads to examine the spatial disparities in freight truck crash severity. Non-freeway roads including local streets and secondary roads are more directly influenced by local traffic patterns, socioeconomic attributes and infrastructure. By excluding freeway incidents, the analysis strengthens the investigation of how these factors contribute to the spatial distribution inequality of freight truck crashes.

The non-freeway road centerlines are obtained from OpenStreetMap by extracting all Ways tagged highway, excluding motorway and

[1] While vehicle weight is a known factor influencing crash severity, detailed weight data for vehicles involved in crashes were unavailable in our dataset. As a proxy, the Collision severity is utilized in the DCI framework, which captures the combined effects of speed, weight, and impact orientation. However, it should be noted that this variable only partially reflects the influence of vehicle weight due to its dependence on multiple factors.

[2] Crash events in the dataset are geocoded with precise latitude and longitude coordinates. To associate each crash point with the underlying road network, a spatial join between the crash coordinates and the OpenStreetMap (OSM) road segments is performed.

motorway link via OSMnx. Crash records are each geocoded in WGS84 and annotated with the crash location field. For each crash not coded as a freeway incident, a nearest neighbor query against the filtered centerlines is performed. To avoid mismatches at overpasses or underpasses, the candidate segments' bridge, tunnel, and layer tags are compared to selecting the segment whose elevation layer and highway classification best match the crash location tag. By excluding crashes on freeways and utilizing this method, the matching errors introduced by elevated or subterranean road segments are avoided.

The prepared crash data is divided into four severity levels: fatal injury, suspected serious injury or severe injury, suspected minor injury or visible injury, and possible injury or complaint of pain. Table 1 shows the distribution of fatalities and injuries in these crashes. The total number of non-freeway freight truck-related crashes is 8728. Among them, there were 262 fatal crashes, resulting in 281 deaths, with the single most severe crash causing 5 deaths. 2106 crashes (24.13 %) resulted in two or more injuries, with the most severe crash causing six or more injuries.

A hotspot analysis of crash locations is demonstrated in Fig. 2, using a kernel density estimation (KDE) to transform geocoded latitude/longitude points into a continuous density surface. Kernel Density tool is applied with search radius (bandwidth) of 1000 m and output cell size of 500 m × 500 m. The heatmap reveals pronounced clustering of freight truck crashes in downtown Los Angeles.

For the independent variables, the direct cause variables of each freight truck-related crash are from the records in the SWITRS database. These variables include driver behavior-related factors, environmental and road conditions, road facilities and control, unsafe speed record, collision severity. The traffic volume data is obtained from the Los Angeles Department of Transportation, which provides detailed and segment-level traffic flow information for major streets and intersections within the city. The data was aggregated at the census block group level to enhance spatial accuracy. In cases where data was missing for certain road segments, the traffic volumes is supplemented based on population density to ensure the dataset coverage. The socioeconomic attributes including mean household income, population density, minority percentage, and employment proportions are extracted from American community survey at the census block group level.[3] The road network pattern indicators were obtained using the OSMnx toolkit for retrieving, modeling, analyzing, and visualizing street networks from OpenStreetMap (OSM) data.

The definitions of variables are demonstrated in Table 2, while the descriptive statistics for discrete/continuous variables are presented in Tables 3 and 4, respectively. All categorical variables were encoded via one-hot encoding and included as inputs to the DCI framework. For instance, for lighting conditions, separate 0/1 dummy variables were created for "daylight," "dawn," "dark with streetlights," and "dark without streetlights." Exactly one of these columns is set to 1 to indicate the observed category at runtime, while the others remain 0. The same procedure was applied to all categorical factors, including weather condition (good, moderate, severe), control device status (functioning, not functioning, absent), and road type (motorway, primary, secondary, other), thereby ensuring that each category is represented explicitly without implying any ordinal relationship.

Table 1
Distribution of freight truck-related crashes by fatalities and injuries.

Number of Deaths/ Injuries per Crash	Fatal crash		Injury crash	
	Number of Crashes	Proportion (%)	Number of Crashes	Proportion (%)
0	8466	97.00	196	2.25
1	248	2.84	6426	73.63
2	12	0.14	1485	17.01
3	0	0.00	410	4.70
4	1	0.01	144	1.65
5	1	0.01	34	0.39
6+	0	0.00	33	0.38
Total	8728	100	8728	100

[3] Specifically, when a crash occurred near or along a highway border, we used a spatial weighting approach to assign the crash to the most relevant census block group, based on the crash's proximity to the tract boundary. In cases where the crash point fell exactly on a boundary, a proportional allocation method is employed to distribute the crash data across the adjacent tracts based on factors such as traffic volume or roadway length.

5. Results

5.1. Distribution of fatal and injury-related freight truck crashes among different socioeconomic, minority, and sector backgrounds

The distribution of freight truck crash severities across different socioeconomic attributes reveals pronounced inequities that underscore the uneven burden borne by vulnerable communities. Table 5 shows that low-income areas experience higher rates of severe crashes, with fatal injuries occurring at 3.921 % and severe injuries at 8.817 %, compared with 2.805 % and 6.061 %, respectively, in high-income areas. These disparities indicate that residents of low-income census block groups endure more serious outcomes when crashes occur. Poor road maintenance, a lack of pedestrian infrastructure, and slower emergency response in these areas increase their risk and perpetuate ongoing spatial inequalities.

An equally concern arises when examining crash severity about neighborhood racial and ethnic composition, as shown in Table 6. Census block groups with a higher proportion of Black or African American residents record a fatal injury rate of 4.281 % and a severe injury rate of 9.606 %, both of which exceed the overall population averages of 3.377 % and 7.212 %. Areas with predominantly Asian residents also exceed the overall average, registering 3.996 % fatal and 8.617 % severe injury rates. These trends show that racially marginalized neighborhoods face higher crash severities, probably due to they are closer to major freight routes, receive less investment in traffic safety infrastructure, and have fewer emergency resources.

Sectoral employment further compounds these inequities as detailed in Table 7. In areas where manufacturing sector employment is highly concentrated, the fatal injury rate rises to 3.880 % and severe injuries to 8.287 %, compared with 2.739 % and 6.229 %, respectively, in low-concentration zones. Similar patterns can also be found in wholesale or retail trade zones. The most significant disparities appear in the transportation and warehousing sector, with high-concentration areas suffering a 4.543 % fatality rate and 9.245 % severe injuries, far above the 2.567 % and 5.695 % observed in low-concentration areas. These patterns indicate that census block groups dominated by freight-related employment experience both greater exposure to heavy freight traffic and insufficient protective measures.

5.2. Individual effects of factors on freight truck crash severity

The distributions of individual treatment effects across all covariates are demonstrated in Fig. 3 and Fig. 4, illustrating how changes in traffic volume, driver behavior, roadway features, and infrastructural variables impact crash severity for single crash occurrences. The average treatment effects estimated from the DCI framework in Table 8 are measured by both the change in severity level and the change in probability of that severity level. Specifically, Traffic volume and driver behavior variables demonstrate the most substantial positive impacts on crash severity. The average severity-level change for annual average daily truck traffic density reaches 0.816, accompanied by a 44.859 % increase in the probability of a more severe outcome. Similarly, the speed dummy yields an average severity-level change of 1.015 and a 48.218 %

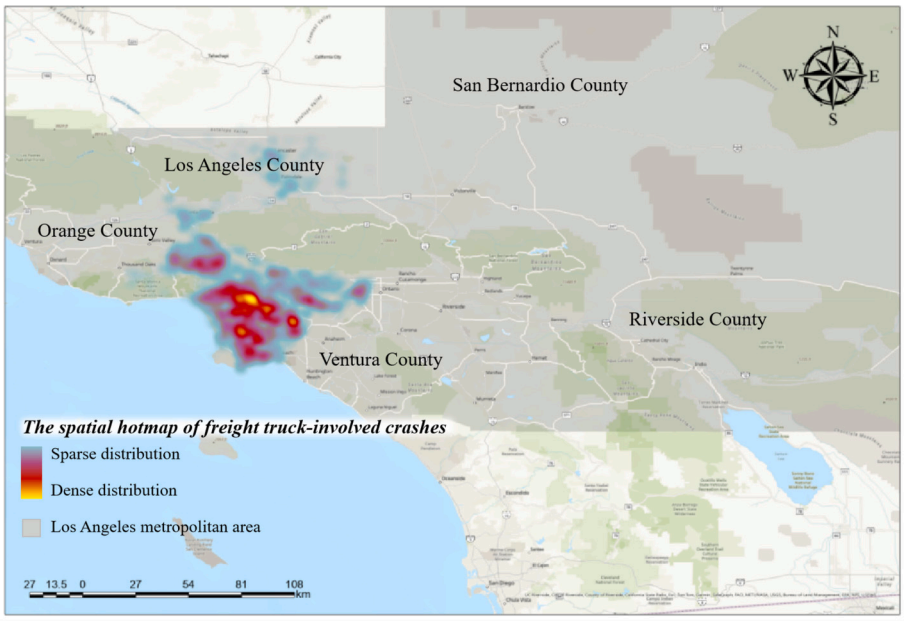


Fig. 2. Spatial hotspot map of the freight truck-involved crashes in Los Angeles metropolitan area.

probability increase. The effect of collision severity is even larger, with an average severity-level change of 1.062 and a corresponding probability change of 58.410 %. These findings indicate that higher speeds, greater truck volumes, and more severe collisions consistently elevate the likelihood of a more serious crash outcome.

Vulnerable user involvement also exerts a pronounced effect, particularly when pedestrians are involved, which produces an average severity-level increase of 0.953 and raises the probability of a more severe outcome by 60.433 %. Bicyclist involvement shows a severity-level change of 0.632 and a 24.418 % probability increase, whereas motorcyclist involvement yields a severity-level change of 0.713 and a 51.771 % increase in probability. The magnitude of these effects underscores the heightened danger posed to vulnerable road users in freight truck crashes, especially pedestrians, whose involvement substantially increases both the severity and the probability of severe injury or fatality. Weather and signal equipment variables present moderate negative effects on crash severity. The adequate lighting conditions reduce the average severity level by 0.315 and lower the probability of a more severe outcome by 18.091 %, while control device condition decreases the average severity level by 0.215 with an 11.842 % probability reduction. Favorable weather conditions yield a negative treatment effect, reducing the severity level by 0.400 and decreasing the probability of more severe outcomes by 26.407 %.

Road condition variables generate smaller negative effects. Lane number exhibits a severity-level change of −0.187 and a probability change of −10.823 %. The average road segment length reduces severity by 0.180 and probability by 9.823 %, while intersection density reduces severity by 0.171 and probability by 8.429 %. These modest negative effects for lane number, road segment length, and intersection density likely reflect enhanced driver vigilance in complex or congested environments, where frequent direction changes and shorter segments prompt drivers to adopt safer behaviors, thereby lowering crash severity. Road network pattern indicators exert relatively small positive effects. PageRank proportion, which captures the relative importance of road network nodes, increases average severity by 0.295 and probability by 17.028 %. Degree centrality, measuring the number of direct connections at a node, shows an average severity increase of 0.236 and a probability increase of 12.403 %. These findings reveal that more central or highly connected road network segments incur slightly higher severity outcomes, perhaps due to the intersections that serve as hubs for freight traffic, concentrate conflicting movements, and magnify the consequences of a crash.

5.3. Counterfactual inference on freight truck-involved crash

The analysis evaluates how various intervention policies alter crash severity across multiple scenarios by applying counterfactual inference and controlled experiments. Fig. 5(a) demonstrates that areas with high population density greater than 4000 people/km^2 exhibit lower crash severity compared to medium- and low-density areas. Under dark conditions without street lighting, crash severity is 0.558 in high-density areas, 0.590 in medium-density areas, and 0.687 in low-density areas, indicating a gradient of increasing severity as population density decreases. When adequate nighttime lighting is present, crash severity decreases to 0.515, 0.556, 0.599 in high-, medium-, and low-density areas, respectively. Adequate nighttime illumination reduces crash severity across all areas, with particular benefits in low-density regions where lighting infrastructure is often inadequate or poorly maintained.

The effect of pedestrian involvement across intersection density levels is illustrated in Fig. 5(b). In areas with higher intersection density, crash severity is lower, a phenomenon that may result from frequent start-and-stop traffic patterns reducing average vehicle speeds. When pedestrians are involved, crash severity measures 0.545 in high-density areas (50.0–150.0 intersections/km^2), 0.604 in medium-density areas, and 0.663 in low-density areas. In the absence of pedestrian involvement, crash severity decreases to 0.376, 0.448, and 0.532 in high-, medium-, and low-density regions, respectively. This phenomenon may be attributed to the frequent stop-and-go traffic in areas with high intersection density, which reduces average vehicle speed and enhances driver attentiveness, thereby resulting in lower crash severity.

Crash severity by control device condition across minority percentage groups is presented in Fig. 5(c). Communities with more than 45 % minority population consistently exhibit higher crash severity. Without control devices, severity is 0.552 in high-minority regions, compared to 0.451 in low-minority and 0.545 in medium-minority regions. When control devices are operational, severity decreases to 0.489 in high-minority areas, 0.487 in medium-minority areas, and 0.445 in low-minority areas. In high-minority communities, the presence of functioning control devices reduces crash severity compared to scenarios lacking such devices, indicating that reliable traffic control can

Table 2
Definitions of dependent and independent variables.

Variable	Type	Definition
Dependent variable		
Crash severity level	Outcome	Severity of freight-truck related crash, ranging from fatal injury to pain.
Independent variable		
Traffic volume		
Lighting condition	Treatment	Lighting conditions during the crash, including daylight, dawn, and dark conditions.
Control device condition	Treatment	Condition of traffic control devices (functioning, not functioning, or absent).
Weather condition	Treatment	The weather conditions at the time of the crash, from good to severe.
Vehicle and driver behavior		
Speed dummy	Treatment	Indicates whether speeding was involved in the crash.
Collision severity	Treatment	Severity of the collision, classified as severe, moderate, or minor.
Improper turning involvement	Treatment	Indicates improper turning involvement in the crash.
Alcohol or drug involvement	Treatment	Indicates alcohol or drug involvement in the crash.
Vulnerable people involvement		
Pedestrian involvement	Treatment	Involvement of pedestrians in the crash.
Bicyclist involvement	Treatment	Involvement of cyclists in the crash.
Motorcyclist involvement	Treatment	Involvement of motorcyclists in the crash.
Road condition		
Lane number	Treatment	Number of lanes in the road segment of the crash location.
Bridge dummy	Treatment	Indicates whether the crash occurred on a bridge.
Tunnel dummy	Treatment	Indicates whether the crash occurred in a tunnel.
Traffic volume		
AADT density	Treatment	The density of the annual average daily traffic volume in the census block group where the crash occurs.
Socioeconomic attributes		
Population density	Confounder	Population density in the census block group where the crash occurs.
Mean household income	Confounder	The mean household income in the census block group where the crash occurs.
Minority attributes		
Black or African American percentage	Confounder	Percentage of the black or African American population in census block groups where crash occurs.
Indian and Alaska Native percentage	Confounder	Percentage of the Indian and Alaska Native population in census block groups where the crash occurs.
Asian percentage	Confounder	Percentage of the Indian and Asian population in census block groups where the crash occurs.
Other minority population percentage	Confounder	Percentage of other minority populations in census block groups where the crash occurs.
Sector attributes		
Manufacturing sector job proportion	Confounder	Proportion of manufacturing sector jobs in the census block group where the crash occurs.
Wholesale trade sector job proportion	Confounder	Proportion of wholesale trade sector jobs in the census block group where the crash occurs.
Retail trade sector job proportion	Confounder	Proportion of retail trade sector jobs in the census block group where the crash occurs.
Transportation and warehousing sector job proportion	Confounder	Proportion of transportation and warehousing sector jobs in the census block group where the crash occurs.
Road network pattern and infrastructure		
PageRank proportion	Confounder	The ratio of the maximum and minimum PageRank values reflects the difference in the distribution of the importance of nodes.
Degree centrality	Confounder	The degree centrality indicators of the road networks measure how central a road segment is in the overall road network.
Cluster coefficient	Confounder	The clustering coefficient indicators of the road network. The higher value reflects the complexity of the road network structure in the area.
Average road segment length	Confounder	The average length of the segments of the road network in census block group where crash occurs.
Intersection density	Confounder	The density of road intersections in census block group where crash occurs.

Table 3
Descriptive statistics for discrete variables utilized in the DCI Framework.

Variable	Value definition	Percentage
Crash severity level	3 = Fatal injury	3.377 %
	2 = Suspected serious injury or severe injury	7.212 %
	1 = Suspected minor injury or visible injury	28.389 %
	0 = Possible injury or complaint of pain	61.023 %
Collision severity	2 = Severe	5.746 %
	1 = Moderate	20.451 %
	0 = Minor	71.737 %
Lighting condition	3 = Daylight	70.838 %
	2 = Dawn	2.935 %
	1 = Dark and streetlights	18.905 %
	0 = Dark and no lights	7.322 %
Control device condition	2 = Functioning	24.381 %
	1 = Not Functioning	0.237 %
	0 = No device	75.383 %
Weather condition	2 = Good weather	86.776 %
	1 = Moderate weather	10.147 %
	0 = Severe weather	3.077 %
Speed dummy	1 = Speeding	34.922 %
	0 = Not speeding	65.078 %
Improper turning involvement	1 = Improper turning involved	15.386 %
	0 = Improper turning not involved	84.614 %
Alcohol or drug involvement	1 = Alcohol or drug involved	5.807 %
	0 = Alcohol or drug not involved	94.193 %
Pedestrian involvement	1 = Pedestrian involved	3.629 %
	0 = Pedestrian not involved	96.371 %
Bicyclist involvement	1 = Cyclist involved	2.036 %
	0 = Cyclist not involved	97.964 %
Motorcyclist involvement	1 = Motorcyclist involved	3.298 %
	0 = Motorcyclist not involved	96.702 %
Road type	2 = Primary	37.415 %
	1 = Secondary	24.933 %
	0 = Other	37.652 %
Lane number	1 = Bidirectional dual lane	23.166 %
	0 = Bidirectional single lane	76.834 %
Bridge dummy	1 = Crash on the bridge	0.915 %
	0 = Crash not on the bridge	99.085 %
Tunnel dummy	1 = Crash in the tunnel	0.047 %
	0 = Crash not in the tunnel	99.953 %

substantially reduce inequities in crash severity.

The influence of weather conditions across mean household income groups is shown in Fig. 5(d), revealing that adverse weather magnifies crash severity most in low-income regions. Under severe weather, crash severity is 0.586 in low-income areas, 0.552 in medium-income areas, and 0.544 in high-income areas. Even under moderate conditions, low-income areas record a crash severity of 0.551 compared to 0.486 in high-income areas, and under good weather, severity remains higher in low-income areas than in high-income areas. These outcomes underscore that weaker infrastructure and less effective traffic management in low-

Table 4
Descriptive statistics for continuous variables utilized in the DCI Framework.

Variable	Type	Unit	Mean	Std	Min	Max
AADT density	Independent	vehicle/km^2	8.163×10^5	7.915×10^5	6.171×10^2	4.344×10^6
Population density	Independent	People/km^2	9.296×10^3	1.397×10^3	0.000	2.109×10^4
Mean household income	Independent	$/year	1.044×10^5	3.292×10^4	1.250×10^4	2.250×10^5
Black or African American percentage	Independent	%	9.196	13.612	0.000	90.922
Indian and Alaska Native percentage	Independent	%	1.914	1.622	0.000	33.333
Asian percentage	Independent	%	16.807	17.621	0.000	100.000
Other minority population percentage	Independent	%	31.884	24.068	0.000	85.900
Manufacturing sector job proportion	Independent	%	8.610	6.905	0.000	100.000
Wholesale trade sector job proportion	Independent	%	3.020	4.045	0.000	38.381
Retail trade sector job proportion	Independent	%	9.616	6.518	0.000	51.190
Transportation and warehousing sector job proportion	Independent	%	5.955	5.911	0.000	56.196
PageRank proportion	Independent	–	17.089	4.797	1.019	34.766
Degree centrality	Independent	–	0.054	0.057	0.007	1.343
Cluster coefficient	Independent	–	0.041	0.028	0.000	0.291
Average road segment length	Independent	m	256.079	338.801	67.163	3029.571
Intersection density	Independent	num/km^2	38.017	22.726	0.027	173.461

Table 5
Crash severity levels statistics grouped by mean household income characteristics.

Mean household income	Fatal injury	Severe injury	Minor injury	Possible injury
Low group	3.921 %	8.817 %	27.982 %	59.280 %
Medium group	3.233 %	6.905 %	26.889 %	62.972 %
High group	2.805 %	6.061 %	27.257 %	63.877 %

Table 6
Crash severity levels statistics grouped by minority characteristics.

Minority population	Fatal injury	Severe injury	Minor injury	Possible injury
Black or African American percentage	4.281 %	9.606 %	27.629 %	58.483 %
Indian and Alaska Native percentage	4.243 %	9.061 %	25.763 %	60.933 %
Asian percentage	3.996 %	8.617 %	25.785 %	61.602 %
Other minority population percentage	4.687 %	9.486 %	27.247 %	58.580 %
All population groups	3.377 %	7.212 %	28.389 %	61.023 %

Table 7
Crash severity levels percentage distribution by sector characteristics.

Sector	Group	Fatal injury	Severe injury	Minor injury	Possible injury
Manufacturing sector job proportion	Low group	2.739 %	6.229 %	29.140 %	61.893 %
	High group	3.880 %	8.287 %	26.199 %	61.633 %
Wholesale trade sector job proportion	Low group	2.805 %	6.061 %	27.257 %	63.877 %
	High group	4.291 %	8.662 %	27.635 %	59.412 %
Retail trade sector job proportion	Low group	2.795 %	6.119 %	29.177 %	61.909 %
	High group	4.147 %	8.480 %	27.583 %	59.791 %
Transportation and warehousing sector job proportion	Low group	2.567 %	5.695 %	28.580 %	63.158 %
	High group	4.543 %	9.245 %	28.356 %	57.857 %

income regions exacerbate the negative impact of adverse weather on crash severity.

6. Discussion

In response to the increasing demand for road freight transportation and the consequent rise in freight truck-related crashes, this study focuses on investigating the spatial equity concerns associated with these accidents. Through a deep counterfactual inference model, our analysis has shown that socio-economic and infrastructural factors contribute significantly to disparities in crash severity across the Los Angeles metropolitan area. Key findings indicate that low-population-density, low-income, and minority-dense areas experience higher crash severity, often due to inadequate road infrastructure and environmental conditions. Our findings provide a detailed analysis of how the aforementioned factors affect freight truck-involved crashes and highlight spatial inequities of minority group residential communities, underscoring the importance of interventions in enhancing both infrastructural and environmental attributes. Our contributions are twofold: First, by applying counterfactual inference, we move beyond traditional factor analysis to quantify the effects of road and environmental conditions on freight truck crash outcomes. Second, our findings highlight the spatial inequities in crash severity across regions, underscoring the demand for targeted safety interventions in vulnerable communities. This research not only deepens the understanding of the factors influencing freight truck crashes but also provides a crucial framework for policymakers to develop more equitable transportation safety strategies.

Our findings enrich the current scholarship in several key directions. We delve into the intricate dynamics of freight truck-related crashes and their spatial equity implications, drawing on established research to bolster our analytical framework. Our deep counterfactual inference model not only builds upon previous work that underscores the significance of socio-economic and infrastructural factors in determining crash severity (Uddin and Huynh, 2017) but also extends their findings by introducing an analysis that quantifies the impact of road and environmental conditions on crash outcomes. This approach allows us to move beyond traditional factor analysis, offering a more nuanced understanding of how specific conditions, such as inadequate lighting and weather-responsive technologies, contribute to the severity of crashes in low-population-density areas. Furthermore, our research complements the work of McDonald et al. (2019) by advocating for resilient road infrastructure programs in low-income areas, emphasizing the importance of resurfacing roads and reinforcing embankments to enhance safety. We also align with Das et al. (2022) in recognizing the need for smart traffic signals in minority-dense communities to ensure the smooth flow of freight trucks, thereby addressing the spatial equity issues that persist despite the presence of traffic control devices. While our

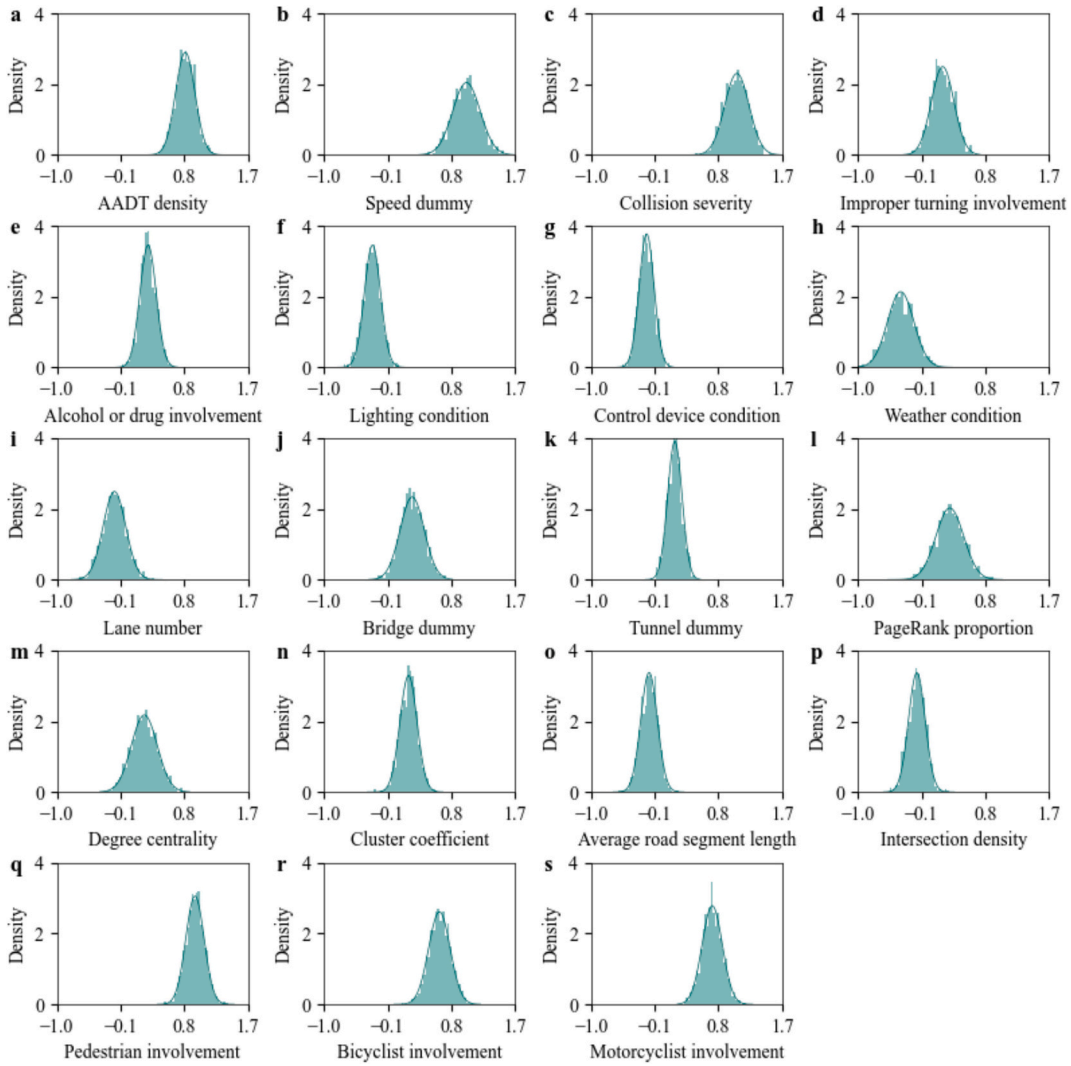


Fig. 3. Distribution of the individual accident severity level among traffic volume, vehicle and driver behavior, weather and signal equipment, road condition, road network pattern, and infrastructure, vulnerable people involvement treatment variables.

study shares the common goal with these esteemed researchers of improving road safety, our methodology provides a distinct advantage by integrating a comprehensive analysis that captures the interplay between various factors influencing crash severity. This deep counterfactual inference model allows us to quantify the effects, offering a more precise understanding of how targeted interventions can lead to more equitable transportation safety strategies. Our findings provide a detailed framework that can guide policymakers in developing targeted and effective interventions to reduce crash severity and promote spatial equity in road freight transportation.

Our study provides far-reaching policy implications for transport and urban planning to reduce the social inequality of safe risks in the less populated regions, low-income communities, and minority concentrations. Firstly, in less populated areas where the freight truck crash severity is high, inadequate road infrastructure increases safety risks. These regions, often rural or suburban, suffer from a lack of essential safety features like lighting, traffic control devices, and proper road maintenance, making them more vulnerable to severe accidents. A key policy intervention is improving lighting along freight corridors and intersections, especially in underserved areas. Installing energy-efficient LED and solar-powered streetlights, equipped with adaptive technologies, can enhance visibility and reduce crash severity (Uddin and Huynh, 2017). Additionally, upgrading traffic control devices is critical. Implementing advanced traffic signal systems with real-time monitoring and adaptive controls can mitigate risks at key intersections with heavy freight traffic (Zhang and Liu, 2020). Traffic calming measures, such as speed bumps and roundabouts, can also slow vehicles and improve safety. Regular infrastructure audits and a comprehensive road safety program will ensure that these areas receive necessary maintenance and upgrades to prevent severe crashes.

Secondly, low-income communities are found to be disproportionately affected by adverse weather, leading to higher freight truck crash severity. These areas often suffer from poor drainage, degraded road surfaces, and a lack of weather-adaptive traffic management tools, making them more vulnerable. To address these issues, policy measures should prioritize resilient infrastructure programs focused on resurfacing roads, reinforcing embankments, and improving traction (McDonald et al., 2019). A broader Climate-Resilient Transportation Plan can target regions prone to extreme weather. Integrating weather-responsive technologies, such as real-time monitoring systems and dynamic speed regulations, can help reduce crash risks by adjusting speed limits based on conditions. Enhanced winter maintenance, including timely snow removal and de-icing, is critical for areas facing severe weather (Shin, 2024a). Freight route optimization systems should also divert trucks from hazardous routes, promoting long-term road safety and resilience in low-income regions.

Thirdly, the counterfactual results reveal a significant spatial equity issue in minority-dense communities, where freight truck crash severity

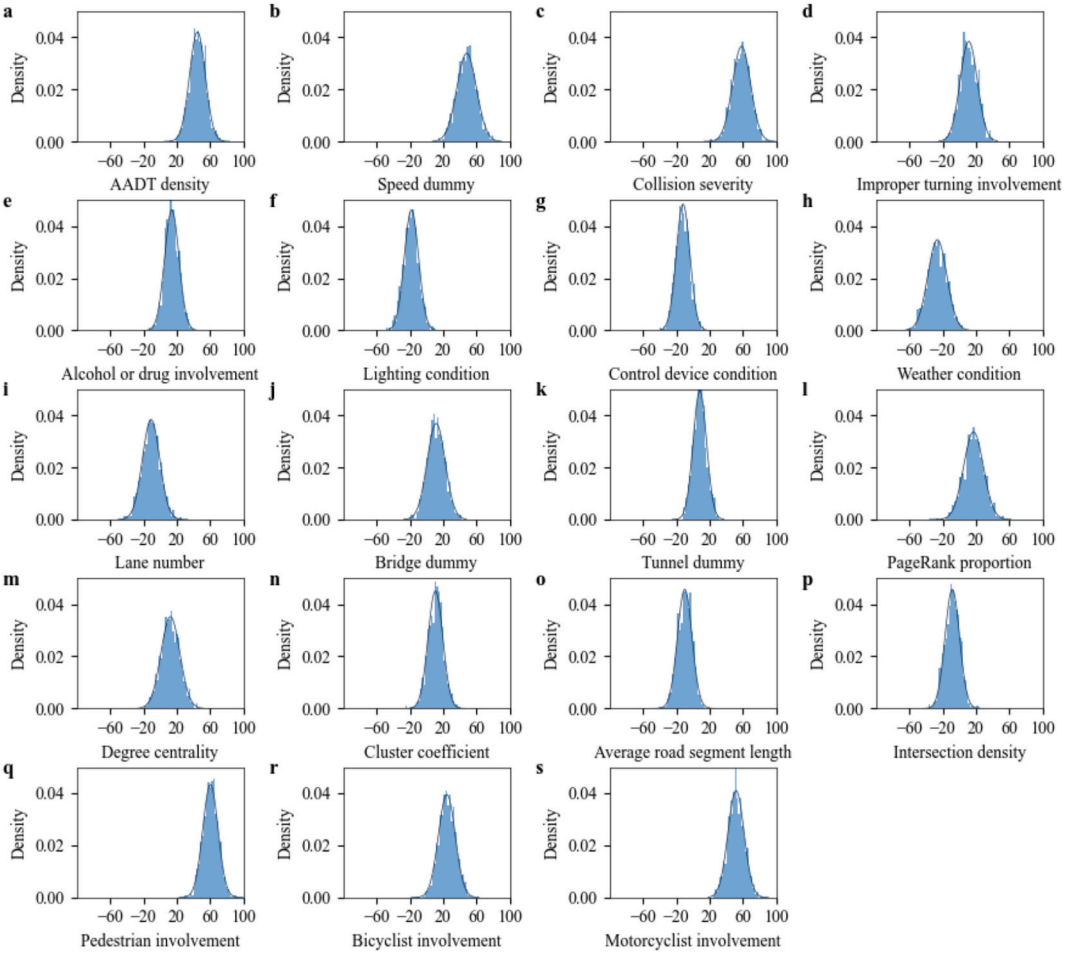


Fig. 4. Distribution of individual accident probability change among traffic volume, vehicle and driver behavior, weather and signal equipment, road condition, road network pattern, infrastructure, and vulnerable people involvement treatment variables.

remains elevated despite the presence of traffic control devices. These areas often suffer from underinvestment in infrastructure and road safety, requiring targeted interventions. Implementing comprehensive traffic management systems, such as smart traffic signals that adapt to real-time conditions, is essential for improving freight truck flow through residential areas (Das et al., 2022). Regular maintenance and upgrades of traffic control devices, including signals, signage, and speed monitoring systems, are crucial for enhancing road safety. Automated enforcement systems, including speed and red-light cameras, can further reduce reckless driving.

Several limitations in our study should be considered to extend the current findings. First, although the deep counterfactual inference model effectively captures various road environmental factors and driver behaviors, the study did not incorporate real-time data. The static nature of the dataset used limits our ability to account for temporal fluctuations in traffic patterns that could influence crash severity. Future research could enhance this analysis by integrating real-time traffic data and temporal factors, providing a more accurate and dynamic understanding of how these conditions impact crash risks. This research primarily addresses spatial equity concerns through infrastructural and socio-economic lenses but does not fully incorporate the complexities of human factors, such as driver fatigue, distraction, or behavioral adaptation to infrastructure. While driver behavior-related variables, such as alcohol or drug involvement, were included, the model does not directly account for more nuanced driver-specific factors that could substantially influence crash outcomes. Future work could explore the integration of behavioral data or telematics-based monitoring to assess how individual driving patterns contribute to the overall severity level in different spatial conditions.

7. Conclusion

This study reveals significant spatial disparities in freight truck crash severity across the Los Angeles metropolitan area, rooted in socio-economic and infrastructural inequities. Using a deep counterfactual inference model, the research highlights critical concerns, such as inadequate lighting, a lack of real-time weather monitoring, and insufficient adaptive traffic management technologies in vulnerable communities. These disparities call for a shift from reactive measures to long-term, sustainable solutions. We advocate for future strategies that integrate real-time data and dynamic modeling to adapt to changing traffic patterns and environmental conditions. Advanced technologies like adaptive lighting and dynamic speed regulation should be paired with regular upgrades of traffic control devices.

However, the study does have several limitations that should be acknowledged. First, there may be inherent biases in the data collection process, including underreporting of certain types of crashes or incidents. In particular, minor crashes, non-injury crashes, or crashes in less well-monitored areas may not be accurately represented in the dataset, potentially skewing the analysis (Islam et al., 2024). Additionally, while the study focuses on infrastructure and environmental factors, driver behavior is a crucial element in determining crash outcomes (Shandhana Rashmi and Marisamynathan, 2023), and the absence of a detailed behavioral analysis could limit the full understanding of the influencing mechanism. Future research should extend the model to other urban areas, particularly those with different infrastructure,

Table 8
Average treatment effect for non-freeway freight truck crash severity represented by average severity level change and average probability change.

Categories	Variables	Average severity level change	Average probability change
Traffic volume	AADT density	0.816	44.859 %
	Speed dummy	1.015	48.218 %
	Collision severity	1.062	58.410 %
Vehicle and driver behavior	Improper turning involvement	0.194	11.295 %
	Alcohol or drug involvement	0.289	14.067 %
	Lighting condition	−0.315	−18.091 %
Weather and signal equipment	Control device condition	−0.215	−11.842 %
	Weather condition	−0.400	−26.407 %
	Lane number	−0.187	−10.823 %
Road condition	Bridge dummy	0.247	11.594 %
	Tunnel dummy	0.184	8.299 %
	PageRank proportion	0.295	17.028 %
	Degree centrality	0.236	12.403 %
Road network pattern and infrastructure	Cluster coefficient	0.195	10.729 %
	Average road segment length	−0.180	−9.823 %
	Intersection density	−0.171	−8.429 %
	Pedestrian involvement	0.953	60.433 %
Vulnerable people involvement	Bicyclist involvement	0.632	24.418 %
	Motorcyclist involvement	0.713	51.771 %

socioeconomic, and geospatial features, to assess the generalizability of the findings. Using weighted averages for ACS data helps reduce some border issues when a crash happens near the edge of a block group. However, this method cannot fully handle cases where a major road divides two very different neighborhoods. In such situations, one crash point on the highway may affect two communities with very different socioeconomic conditions, and averaging their data can hide the real local situation. Future research should explore other methods, such as using detailed land-use buffers or categorizing neighborhoods qualitatively, to better capture these sharp divides along major roads. Integrating behavioral factors such as driver fatigue, distractions, and decision-making processes could provide a more holistic view of the causes of freight truck crashes. Further exploration of these aspects, perhaps through the use of more granular telematics data or behavioral simulations, would enrich the understanding of the interactions between infrastructure, driver behavior, and crash outcomes.

The model and analytical farmwork implemented in our study are highly repeatable, reproducible, and replicable in a broader context for transportation safety. By implementing data-driven and targeted interventions, our study provides the possibility for policymakers to create a safer, more equitable road system, ensuring that all residents, regardless of socioeconomic status, are prioritized in road safety efforts.

CRediT authorship contribution statement

Yichen Wang: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Hao Yin:** Resources, Investigation. **Yifan Yang:** Writing – review & editing, Software. **Chenyang Zhao:** Validation, Software. **Siqin Wang:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

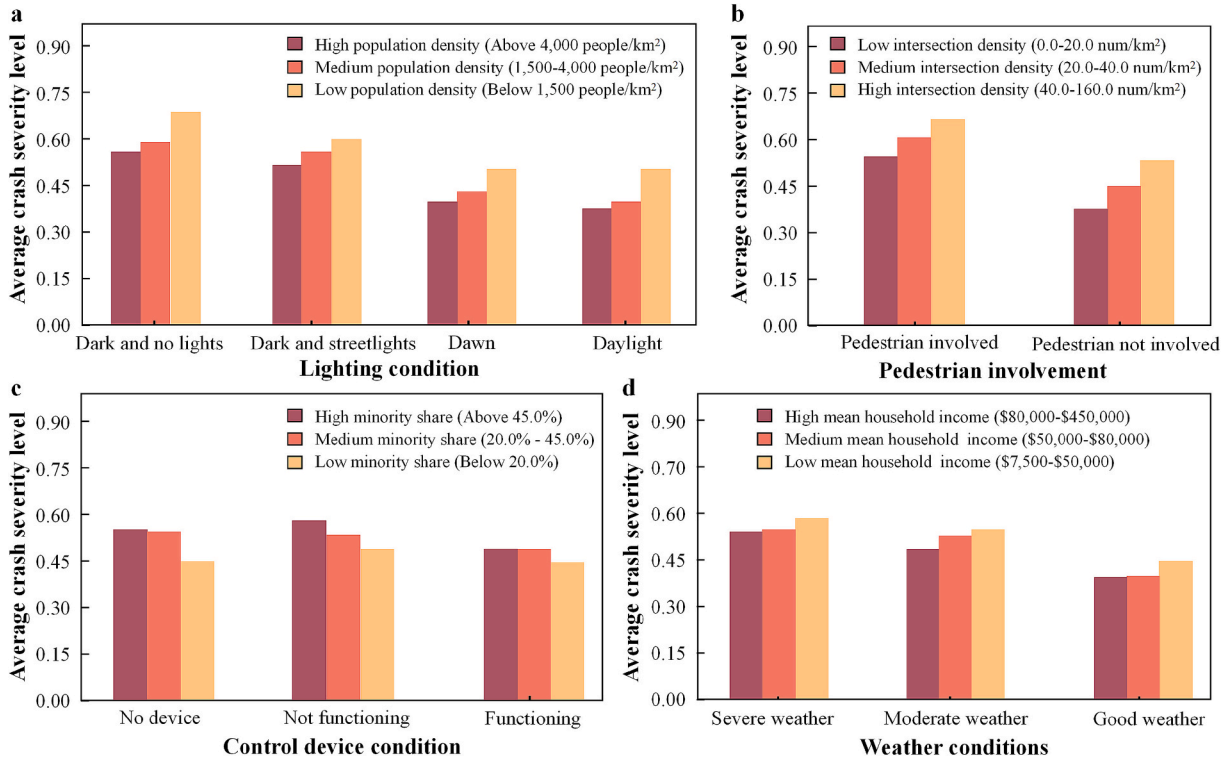


Fig. 5. Counterfactual inference result. (a) Impact of lighting conditions among different population density groups. (b) Impact of pedestrian involvement among different intersection density groups. (c) Impact of control device condition among different minority share groups. (d) Impact of weather conditions among different mean household income groups.

Appendix

A.1. The Deep counterfactual inference (DCI) framework

The input to the model includes observed variables and preliminary counterfactual inference results, denoted as $\left[\mathbf{T}_i, \mathbf{T}_i^*, \mathbf{X}_i, \mathbf{U}_i\right], \forall i \in I$. Considering the complexity of factors causing freight truck-related crashes (Shinar and Hauer, 2024; Yan et al., 2024), which include binary variables and ordinal variables that directly represent immediate causes, as well as environmental factors represented by continuous variables (Haddad et al., 2023; Hong et al., 2019), a calibration component is employed in the model. This component helps to map continuous variables, binary variables, and ordinal variables to a standardized range to ensure consistency and facilitate learning. Specifically, each categorical variable is treated as a separate input, and one-hot encoding is applied to represent all its categories appropriately, and the continuous variables are scaled to a standardized range (e.g., [0,1]) using min-max normalization, see Eq. (A1).

$$\widehat{x}_i = \frac{x_i - \min(x)}{\max(x) - \min(x)}, \forall x_i \in \mathbf{X}_i \cup \mathbf{T}_i \tag{A 1}$$

The shared representation component comprises a series of fully connected layers that learn a unified high-level embedding of the calibrated inputs, enabling the model to integrate diverse factors and capture complex, nonlinear patterns essential for accurate prediction. Because interaction effects among causes are consistent across factual and counterfactual scenarios, these layers are shared between both tasks; the network subsequently branches into two task-specific modules. The resulting latent embedding from the shared layers is given by Eq. (A2).

$$H_i = \phi(W_2 \times \phi(W_1 \times \widehat{X}_i + b_1) + b_2) \tag{A 2}$$

where $\widehat{X}_i$ denotes the concatenated calibrated input vector; W_1, W_2 are weight matrices and b_1, b_2 represent bias vectors; $\phi(\cdot)$ is the activation function. After obtaining the shared representation H_i, the model branches into task-specific modules. The factual outcome head predicts the probability distribution over k severity levels under the observed conditions while the counterfactual outcome head predicts the probability distribution under alternative scenarios, as Eq. (A3) and Eq. (A4) respectively.

$$\widehat{\mathbf{Y}}_i^F = \text{softmax}(f_{Factual}(H_i)) \in \mathbb{R}^k \tag{A 3}$$

$$\widehat{\mathbf{Y}}_i^{CF} = \text{softmax}\left(f_{Counterfactual}(H_i)\right) \in \mathbb{R}^k \tag{A 4}$$

where $f_{Factual}, f_{Counterfactual}$ are neural network functions, similar as fully connected layers that output logits for each of the k severity levels. The SoftMax function ensures that the outputs are valid probability distributions over the k classes.

A sparse spatial weights matrix at the census block group level is constructed to explicitly account for spatial autocorrelation among crash observations. The centroid of each census block group within the study area is identified, and the shortest-path distances between all pairs of centroids along the existing road network are calculated rather than using straight-line Euclidean distances to capture realistic connectivity in freight corridors. For each origin census block group, these distances are converted into weights via an inverse distance function. The normalized spatial weight vectors are included as additional inputs to the calibration component of our deep counterfactual inference network.

A composite loss function combining prediction errors and regularization terms is defined to address latent confounders.

$$Loss_{DCI} = Loss_{Factual} + \gamma \times Loss_{Counterfactual} + \lambda \times Loss_{Reg} \tag{A 5}$$

where the loss function of the factual scenario of the freight truck-related crashes $Loss_{Factual}$ is written as Eq. (A6) using categorical cross-entropy loss. $Loss_{Counterfactual}$ In Eq. (A7) measures the error in predicting the preliminary counterfactual estimates Y_i^* using categorical cross-entropy loss as well. And $Loss_{Reg}$ is designed to mitigate the influence of unobserved confounders U_i. Hyperparameters λ, γ are employed to control the importance of the counterfactual loss and regularization term, respectively.

$$Loss_{Factual} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k} \times log\left(\widehat{y_{i,k}^F}\right) \tag{A 6}$$

$$Loss_{Counterfactual} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k}^* \times log\left(\widehat{y_{i,k}^{CF}}\right) \tag{A 7}$$

$$Loss_{Reg} = \|\boldsymbol{z}\|_2^2 \tag{A 8}$$

Where $\boldsymbol{z}$ is the latent representation vector that encodes the unobserved confounders U, $\|\boldsymbol{z}\|_2^2$ is the squared L_2-norm (which represents the magnitude of the latent representation), used to regularize the network by penalizing large values in the latent variables. The pseudocode for deep counterfactual inference is presented in Table A1.

Table A1
Pseudocode for deep counterfactual inference algorithms.

Input: Dataset D = (X_i, T_i, Y_i, T'$_i$, Y'$_i$, U_i) for $i = 1$ to N, hyperparameters λ_1, λ_2, λ_{reg}, α
01. Initialize model parameters θ
02. Compute propensity scores P(T_i | X_i) for each i using Propensity Score Matching (PSM)
03. **For** each epoch:
04. **For** each batch B ⊆ D:
05. **For** each sample i in batch B:
06. // Step 1: Calibration
07. Apply calibration to covariates X_i and confounders Z_i to standardize variables
08. X_i = Calibrate(X_i, Z_i)
09. // Step 2: Generate latent variable U
10. U_i = Sample U ~ N(0,1) or other distribution
11. // Step 3: Forward Pass
12. Compute shared representation Hi via shared layers
13. H_i = Shared_Layers([X_i, U_i])
14. Predict factual outcome $\hat{Y}_i^F$ using factual outcome head
15. $\hat{Y}_i^F$ = Factual_Head(H_i, T_i)
16. Predict counterfactual outcome $\hat{Y}_i^{CF}$ using counterfactual outcome head
17. $\hat{Y}_i^{CF}$ = Counterfactual_Head(H_i, T'$_i$)
18. // Step 4: Loss Computation
19. Compute factual loss LF using cross-entropy loss with weight λ_1
20. LF = λ_1 × CrossEntropy(Y_i, $\hat{Y}_i^F$)
21. Compute counterfactual loss LCF using cross-entropy loss with weight λ_2
22. LCF = λ_2 × CrossEntropy(Y'$_i$, $\hat{Y}_i^{CF}$)
23. Compute regularization term LReg with weight λ_{reg} to account for latent variable U
24. $L_{Reg} = \lambda_{reg} \times ||\theta||^2$
25. Compute the total loss
26. $L = L^F + L^{CF} + L_{Reg}$
27. // Step 5: Backward Pass
28. Compute gradients ∇θL
29. // Step 6: Update Parameters
30. Update model parameters θ using Adam optimizer
31. θ = θ - α × ∇θL
32. **Output:** Trained model capable of estimating Individual Treatment Effect (ITE) and Average Treatment Effect (ATE)
Function Definitions:
33. Function Calibrate(X, Z):
34. // Standardizes covariates X and confounders Z using predefined calibration functions
35. **Return** calibrated [X, Z]
36. Function CrossEntropy(y_true, y_pred):
37. // Computes cross-entropy loss between the true outcome and predicted outcome
38. **Return** loss_value
39. Function PropensityScoreMatching(X, T):
40. // Generates propensity scores for given covariates X and treatment T using logistic regression or machine learning models
41. **Return** propensity_scores
42. Function AdamOptimizer(gradients, learning_rate):
43. // Updates model parameters using the Adam optimization algorithm
44. **Return** updated_parameters

Appendix B Model configuration and parameter settings.

The parameter configuration of the deep counterfactual inference model is shown in Table A2. The model's architecture is structured with shared layers, followed by two distinct branches as one handling factual predictions and the other dedicated to counterfactual outcomes. Both branches are configured identically, each comprising 256 hidden units to facilitate robust feature processing. This setup ensures that the model can effectively capture the complex patterns present in the crash data, which are often shaped by a combination of weather conditions, driver behavior, and road characteristics. The shared layers, composed of several dense layers with 128 neurons per layer, play a critical role in learning the intricate relationships between these input factors. The use of the ReLU activation function further enhances the model's ability to handle non-linearity, while simultaneously addressing the gradient vanishing problem. To address the issue of overfitting, a dropout rate of 0.3 is implemented, where a portion of neurons are randomly deactivated during each training iteration. This process forces the network to generalize better, ensuring that it does not become overly reliant on any specific set of features. Furthermore, the model employs He initialization, which maintains the stability of gradients across layers, thus ensuring more consistent learning.

The model's loss function incorporates both factual and counterfactual loss components. The factual prediction loss is given greater importance with a weight of $\lambda_1 = 0.65$, emphasizing the need for accurate predictions in observed data. In contrast, the counterfactual ordering loss which captures the rank order of outcomes in hypothetical scenarios, is assigned a weight of $\lambda = 0.35$. This balance ensures that while prediction accuracy remains the priority, the model also effectively captures the nuances of alternative scenarios. Moreover, the regularization with a coefficient of $\lambda_{reg} = 0.01$ is applied to manage model complexity, reducing the risk of overfitting by penalizing large weights. We use the Adam optimizer, which dynamically adjusts the learning rate based on the first and second moments of the gradients. With an initial learning rate $\alpha = 0.001$, the model begins with relatively large updates, which gradually decrease as the training progresses, enabling fine-tuning. The batch size is set at 64, and the model is trained over 200 epochs, allowing sufficient time for convergence. An early stopping mechanism is employed to mitigate the risk of overfitting, ensuring that training halts when validation performance stabilizes.

The model accounts for unobserved confounding factors by incorporating latent variables *U*. These variables are designed to capture the influence

of factors that are not directly observed but still impact crash severity outcomes. The latent variables are modeled with a dimension of $U_d = 10$ and are assumed to follow a normal distribution $U \sim N(0, 1)$. This allows the model to incorporate hidden factors, thereby improving the robustness and accuracy of its inferences, both for factual and counterfactual scenarios.

Table A2
Parameter configuration of the deep counterfactual inference model.

Parameter Name	Description	Value
Number of Dense Layers (L)	Number of fully connected layers in the shared representation component of the network.	4
Neurons numbers (N)	Number of neurons in each dense layer, representing model capacity to capture data patterns.	128
Dropout rate (D)	Probability of deactivating neurons during training to reduce overfitting.	0.3
Hidden units(H)	Number of hidden units in factual and counterfactual branches for processing predictions.	256
Output neurons (O)	Neurons correspond to the four crash severity levels (fatal, severe, minor, and possible injuries).	4
Regularization (L_2)	L_2 regularization applied to task-specific layers to penalize large weight magnitudes.	0.01
Learning rate (α)	Initial step size for gradient updates, decayed over time for stable convergence.	0.001
Batch size (B)	Number of samples processed before model weight updates.	64
Epochs (E)	Number of complete passes through the training dataset.	200
Factual loss weight (λ_1)	Weight balancing the contribution of the factual scenario to the overall loss function (prediction loss).	0.65
Counterfactual ordering loss weight (λ_2)	Weight emphasizes the importance of counterfactual ordering loss during training.	0.35
Regularization weight (λ_{reg})	Regularization weight controlling the penalty applied to model weights.	0.01
Latent variable dimension (U_d)	Latent variable dimension representing unobserved factors influencing crashes.	10

Appendix C: Prediction performance in factual and counterfactual scenarios

To assess the prediction performance of the proposed deep counterfactual inference model, a comparison was made with three alternative methodologies: Causal Forest (CF), Double Machine Learning (DML), and Generative Adversarial Networks (GANs). Causal Forest employs an ensemble of decision trees to estimate individual treatment effects via recursive partitioning (Wager and Athey, 2018). It divides the covariate space into subgroups, where the relationship between covariates and treatment effects can be considered homogeneous. For each partition, the average treatment effect is calculated based on leaf-level observations. The model captures the conditional average treatment effect by aggregating treatment effect estimates across multiple trees, enabling non-parametric and flexible handling of heterogeneous effects.

Double Machine Learning aims to estimate low-dimensional parameters while controlling for high-dimensional nuisance parameters through regularization (Chernozhukov et al., 2018). By applying regularization methods, DML addresses the complexity introduced by these nuisance parameters. To reduce the estimation bias, the DML further incorporates Neyman-orthogonal scores which ensure reduced sensitivity to errors in the nuisance parameters.

Generative Adversarial Networks (GANs)-based counterfactual inference uses a generator–discriminator framework (Yoon et al., 2018). The generator is responsible for producing counterfactual outcomes under varying treatment conditions, while the discriminator evaluates whether the generated outcomes are plausible within the given data context. Rather than matching observed outcomes directly, the generator's objective is to produce counterfactuals consistent with the distribution implied by a baseline model. The process is framed as a minimax optimization problem, where the generator seeks to minimize the discrepancy between the generated counterfactual outcomes and the expected outcomes under the reference model, and the discriminator maximizes this discrepancy. Through adversarial training, the generator iteratively improves its counterfactual predictions while preserving consistency with underlying relationships and data structure.

The performance metrics comparison for crash severity level prediction in both factual and counterfactual scenarios in Table A3 demonstrates the superiority of the deep counterfactual inference model. Additionally, the performance results revealed a consistent trend across all four methods: each model performed better in predicting factual accident severity levels compared to counterfactual ones. It suggests that learning from observed accident data is inherently more straightforward, as factual labels represent real-world outcomes, whereas counterfactual labels involve the simulation of alternative scenarios. Despite the robustness of the deep counterfactual inference model in generating and analyzing counterfactual scenarios, it achieves higher accuracy in predicting factual outcomes.

Table A3
Performance metrics comparison for crash severity level prediction in factual and counterfactual scenarios.

Model	Scenario	MSE	RMSE	MAE
Causal Forest	Factual	1.962	1.394	0.933
Double Machine Learning	Factual	1.325	1.140	0.630
Generative Adversarial Networks	Factual	1.029	1.001	0.475
Our Method	Factual	0.874	0.929	0.401
Causal Forest	Counterfactual	2.091	1.439	0.987
Double Machine Learning	Counterfactual	1.511	1.213	0.704
Generative Adversarial Networks	Counterfactual	1.105	1.038	0.520
Our Method	Counterfactual	0.964	0.980	0.459

Data availability

Data will be made available on request.

References

Ahmed, M.M., Franke, R., Ksaibati, K., Shinstine, D.S., 2018. Effects of truck traffic on crash injury severity on rural highways in Wyoming using Bayesian binary logit models. Accid. Anal. Prev. 117, 106–113. https://doi.org/10.1016/j.aap.2018.04.011.

Al-Bdairi, N.S.S., Hernandez, S., 2020. Comparison of contributing factors for injury severity of large truck drivers in run-off-road crashes on rural and urban roadways: accounting for unobserved heterogeneity. Int. J. Transp. Sci. Technol. 9 (2), 116–127. https://doi.org/10.1016/j.ijtst.2020.01.004.

Bassan, S., 2016. Overview of traffic safety aspects and design in road tunnels. IATSS Res. 40 (1), 35–46. https://doi.org/10.1016/j.iatssr.2016.02.002.

Becker, N., Rust, H.W., Ulbrich, U., 2022. Weather impacts on various types of road crashes: a quantitative analysis using generalized additive models. Eur. Transp. Res. Rev. 14 (1), 37.

Choudhary, P., Imprialou, M., Velaga, N.R., Choudhary, A., 2018. Impacts of speed variations on freeway crashes by severity and vehicle type. Accid. Anal. Prev. 121, 213–222.

Chernozhukov, V., Nekipelov, D.N., Semenova, V., Syrgkanis, V., 2018. Plug-in regularized estimation of high-dimensional parameters in nonlinear semiparametric. models (No. CWP41/18). cemmap Work Pap.

Das, D., Altekar, N.V., Head, K.L., Saleem, F., 2022. Traffic signal priority control strategy for connected emergency vehicles with dilemma zone protection for freight vehicles. Transp. Res. Rec. 2676 (1), 499–517. https://doi.org/10.1177/03611981211039157.

Dumbaugh, E., Rae, R., 2009. Safe urban form: revisiting the relationship between community design and traffic safety. J. Am. Plan. Assoc. 75 (3), 309–329. https://doi.org/10.1080/01944360902950349.

Duvvuri, S., Pulugurtha, S.S., Mathew, S., 2022. Modeling injury severity of crashes involving trucks: capturing and exploring risk factors associated with land use and demographic in addition to crash, driver, and on-network characteristics. IATSS Res. 46 (4), 602–613. https://doi.org/10.1016/j.iatssr.2022.11.001.

Elvik, R., 2024. Risk factors as causes of accidents: criterion of causality, logical structure of relationship to accidents and completeness of explanations. Accid. Anal. Prev. 197, 107469. https://doi.org/10.1016/j.aap.2024.107469.

Faisal Habib, M., Motuba, D., Huang, Y., 2024. Beyond the surface: exploring the temporally stable factors influencing injury severities in large-truck crashes using mixed logit models. Accid. Anal. Prev. 205, 107650. https://doi.org/10.1016/j.aap.2024.107650.

FMCSA, 2021. Large Truck and Bus Crash Facts 2021. https://www.fmcsa.dot.gov/safety/data-and-statistics/large-truck-and-bus-crash-facts-2021.

Fountas, G., Fonzone, A., Gharavi, N., Rye, T., 2020. The joint effect of weather and lighting conditions on injury severities of single-vehicle accidents. Anal. Methods Accid. Res. 27, 100124. https://doi.org/10.1016/j.amar.2020.100124.

Guo, Q., Xu, P., Pei, X., Wong, S.C., Yao, D., 2017. The effect of road network patterns on pedestrian safety: a zone-based Bayesian spatial modeling approach. Accid. Anal. Prev. 99, 114–124. https://doi.org/10.1016/j.aap.2016.11.002.

Haddad, A.J., Mondal, A., Bhat, C.R., Zhang, A., Liao, M.C., Macias, L.J., Watkins, S.C., 2023. Pedestrian crash frequency: unpacking the effects of contributing factors and racial disparities. Accid. Anal. Prev. 182, 106954. https://doi.org/10.1016/j.aap.2023.106954.

Hartford, J., Lewis, G., Leyton-Brown, K., Taddy, M., 2017. Deep IV: A flexible approach for counterfactual prediction. In: Paper presented at the International Conference on Machine Learning.

Hong, J., Park, J., Lee, G., Park, D., 2019. Endogenous commercial drivers' traffic violations and freight truck-involved crashes on mainlines of expressway. Accid. Anal. Prev. 131, 327–335.

Hossain, A., Sun, X., Islam, S., Alam, S., Mahmud Hossain, M., 2023. Identifying roadway departure crash patterns on rural two-lane highways under different lighting conditions: association knowledge using data mining approach. J. Saf. Res. 85, 52–65. https://doi.org/10.1016/j.jsr.2023.01.006.

Hossain, A., Sun, X., Das, S., Jafari, M., Rahman, A., 2024a. Investigating pedestrian-vehicle crashes on interstate highways: applying random parameter binary logit model with heterogeneity in means. Accid. Anal. Prev. 199, 107503. https://doi.org/10.1016/j.aap.2024.107503.

Hossain, A., Sun, X., Mahmud Zafri, N., Codjoe, J., 2024b. Investigating pedestrian crash patterns at high-speed intersection and road segments: findings from the unsupervised learning algorithm. Int. J. Transp. Sci. Technol. 14, 186–201. https://doi.org/10.1016/j.ijtst.2023.04.007.

Islam, M.R., Wang, D., Abdel-Aty, M., 2024. Calibrated confidence learning for large-scale real-time crash and severity prediction. npj Sustain. Mobil. Transp. 1 (1), 1. https://doi.org/10.1038/s44333-024-00001-9.

Jain, A., van der Heijden, R., Marchau, V., Bruckmann, D., 2020. Towards rail-road online exchange platforms in EU-freight transportation markets: an analysis of matching supply and demand in multimodal services. Sustainability 12 (24), 10321.

Johansson, F., Shalit, U., Sontag, D., 2016. Learning representations for counterfactual inference. In: Paper Presented at the International Conference on Machine Learning.

Kim, J., Saleem Saad Al-Bdairi, N., Hernandez, S., 2024. Lane numbers and their impact on commercial motor vehicle crash safety: an econometric perspective. Int. J. Transp. Sci. Technol. https://doi.org/10.1016/j.ijtst.2024.01.002.

Li, J., Liu, J., Liu, P., Qi, Y., 2020. Analysis of factors contributing to the severity of large truck crashes. Entropy 22 (11), 1191.

Lord, D., Washington, S.P., Ivan, J.N., 2005. Poisson, Poisson-gamma and zero-inflated regression models of motor vehicle crashes: balancing statistical fit and theory. Accid. Anal. Prev. 37 (1), 35–46. https://doi.org/10.1016/j.aap.2004.02.004.

Madarshahian, M., Hawkins, J., Huynh, N., Siddiqui, C.K.A., 2024. Investigation of factors affecting crash severity of rear-end crashes with high collision speeds in work zones: a South Carolina case study. Int. J. Transp. Sci. Technol. https://doi.org/10.1016/j.ijtst.2024.07.003.

McDonald, N., Yuan, Q., Naumann, R., 2019. Urban freight and road safety in the era of e-commerce. Traffic Inj. Prev. 20 (7), 764–770.

Mehdizadeh, M., Shariat-Mohaymany, A., Nordfjaern, T., 2019. Driver behaviour and crash involvement among professional taxi and truck drivers: light passenger cars versus heavy goods vehicles. Transport. Res. F: Traffic Psychol. Behav. 62, 86–98.

Naik, B., Tung, L.-W., Zhao, S., Khattak, A., 2016. Weather impacts on single-vehicle truck crash injury severity. J. Saf. Res. 58, 57–65.

Najaf, P., Thill, J.-C., Zhang, W., Fields, M.G., 2018. City-level urban form and traffic safety: a structural equation modeling analysis of direct and indirect effects. J. Transp. Geogr. 69, 257–270. https://doi.org/10.1016/j.jtrangeo.2018.05.003.

Newnam, S., Goode, N., 2015. Do not blame the driver: a systems analysis of the causes of road freight crashes. Accid. Anal. Prev. 76, 141–151. https://doi.org/10.1016/j.aap.2015.01.016.

Newnam, S., Blower, D., Molnar, L., Eby, D., Koppel, S., 2018. Exploring crash characteristics and injury outcomes among older truck drivers: an analysis of truck-involved crash data in the United States. Saf. Sci. 106, 140–145.

NHTSA, 2021. Traffic safety facts 2021: a compilation of motor vehicle traffic crash data (report no. DOT HS 813 527).

Okafor, S., Adanu, E.K., Jones, S., 2022. Severity analysis of crashes involving in-state and out-of-state large truck drivers in Alabama: a random parameter multinomial logit model with heterogeneity in means and variances. Heliyon 8 (12).

Park, S., Park, J., 2022. Multilevel mixed-effects models to identify contributing factors on freight vehicle crash severity. Sustainability 14 (19). https://doi.org/10.3390/su141911804.

Pawlowski, N., Coelho de Castro, D., Glocker, B., 2020. Deep structural causal models for tractable counterfactual inference. Adv. Neural Inf. Proces. Syst. 33, 857–869.

Pew, T., Warr, R.L., Schultz, G.G., Heaton, M., 2020. Justification for considering zero-inflated models in crash frequency analysis. Transp. Res. Interdisc. Perspect. 8, 100249. https://doi.org/10.1016/j.trip.2020.100249.

Rahimi, A., Azimi, G., Asgari, H., Jin, X., 2019. Clustering approach toward large truck crash analysis. Transp. Res. Rec. 2673 (8), 73–85. https://doi.org/10.1177/0361198119839347.

Ramiani, M.B., Shirazian, G., 2020. Ranking and determining the factors affecting the road freight accidents model. Civ. Eng. J. 6 (5), 928–944.

Rashmi, B.S., Marisamynathan, S., 2023. Factors affecting truck driver behavior on a road safety context: a critical systematic review of the evidence. J. Traffic Transp. Eng. 10 (5), 835–865.

Saha, S., Schramm, P., Nolan, A., Hess, J., 2016. Adverse weather conditions and fatal motor vehicle crashes in the United States, 1994-2012. Environ. Health 15 (1), 104. https://doi.org/10.1186/s12940-016-0189-x.

Shin, E.J., 2024a. Factors associated with different types of freight crashes: a macro-level analysis. J. Saf. Res. 88, 244–260.

Shin, E.J., 2024b. Patterns and sources of spatial inequity in freight crashes: an application of decomposition analysis. Accid. Anal. Prev. 205, 107683. https://doi.org/10.1016/j.aap.2024.107683.

Shinar, D., Hauer, E., 2024. Crash causation, countermeasures, and policy – editorial. Accid. Anal. Prev. 201, 107543. https://doi.org/10.1016/j.aap.2024.107543.

Song, L., Fan, W., 2021. Exploring truck driver-injury severity at intersections considering heterogeneity in latent classes: a case study of North Carolina. Int. J. Transp. Sci. Technol. 10 (2), 110–120. https://doi.org/10.1016/j.ijtst.2020.12.006.

Sze, N., Song, Z., 2019. Factors contributing to injury severity in work zone related crashes in New Zealand. Int. J. Sustain. Transp. 13 (2), 148–154.

Taylor, S.G., Russo, B.J., James, E., 2018. A comparative analysis of factors affecting the frequency and severity of freight-involved and non-freight crashes on a major freight corridor freeway. Transp. Res. Rec. 2672 (34), 49–62.

Uddin, M., Huynh, N., 2017. Truck-involved crashes injury severity analysis for different lighting conditions on rural and urban roadways. Accid. Anal. Prev. 108, 44–55. https://doi.org/10.1016/j.aap.2017.08.009.

Ulak, M.B., Ozguven, E.E., 2024. Identifying the latent relationships between factors associated with traffic crashes through graphical models. Accid. Anal. Prev. 197, 107470. https://doi.org/10.1016/j.aap.2024.107470.

Wager, S., Athey, S., 2018. Estimation and inference of heterogeneous treatment effects using random forests. J. Am. Stat. Assoc. 113 (523), 1228–1242.

Wang, C., Chen, F., Zhang, Y., Cheng, J., 2022a. Spatiotemporal instability analysis of injury severities in truck-involved and non-truck-involved crashes. Anal. Methods Accid. Res. 34, 100214. https://doi.org/10.1016/j.amar.2022.100214.

Wang, H., Han, J., Su, M., Wan, S., Zhang, Z., 2021. The relationship between freight transport and economic development: a case study of China. Res. Transp. Econ. 85, 100885.

Wang, J., Lu, H., Sun, Z., Wang, T., 2020. Exploring factors influencing injury severity of vehicle at-fault accidents: a comparative analysis of passenger and freight vehicles. Int. J. Environ. Res. Public Health 17 (4), 1146.

Wang, J., Parajuli, S., Cherry, C.R., McDonald, N.C., Lyons, T., 2022b. Vulnerable road user safety and freight vehicles: a case study in North Carolina and Tennessee. Transp. Res. Interdisc. Perspect. 15, 100650. https://doi.org/10.1016/j.trip.2022.100650.

Wei, F., Dong, D., Liu, P., Guo, Y., Wang, Z., Li, Q., 2022. Quarterly instability analysis of injury severities in truck crashes. Sustainability 14 (21), 14055.

Yan, H., Liao, Y., Ma, Z., Ma, X., 2024. Improving multi-modal transportation recommendation systems through contrastive De-biased heterogenous graph neural networks. Transp. Res. Part C: Emerg. Technol. 164, 104689. https://doi.org/10.1016/j.trc.2024.104689.

Yang, C., Chen, M., Yuan, Q., 2021a. The application of XGBoost and SHAP to examining the factors in freight truck-related crashes: an exploratory analysis. Accid. Anal. Prev. 158, 106153. https://doi.org/10.1016/j.aap.2021.106153.

Yang, C., Chen, M., Yuan, Q., 2021b. The geography of freight-related accidents in the era of E-commerce: evidence from the Los Angeles metropolitan area. J. Transp. Geogr. 92, 102989. https://doi.org/10.1016/j.jtrangeo.2021.102989.

Yoon, J., Jordon, J., Van Der Schaar, M., 2018. GANITE: Estimation of individualized treatment effects using generative adversarial nets. In: Paper Presented at the International Conference on Learning Representations.

Yu, C., Hua, W., Yang, C., Fang, S., Li, Y., Yuan, Q., 2024. From sky to road: incorporating the satellite imagery into analysis of freight truck-related crash factors. Accid. Anal. Prev. 200, 107491. https://doi.org/10.1016/j.aap.2024.107491.

Yu, C., Dong, W., Lin, H., Lu, Y., Wan, C., Yin, Y., Yuan, Q., 2025. Multi-layer regional railway network and equitable economic development of megaregions. npj Sustain. Mobil. Transp. 2 (1), 3. https://doi.org/10.1038/s44333-024-00024-2.

Yuan, Q., Wang, J., 2021. Goods movement, road safety, and spatial inequity: evaluating freight-related crashes in low-income or minority neighborhoods. J. Transp. Geogr. 96, 103186.

Yuan, Q., Lin, H., Yu, C., Yang, C., 2023. Modeling freight truck-related traffic crash hazards with uncertainties: a framework of interpretable Bayesian neural network with stochastic variational inference. Int. J. Transp. Sci. Technol. https://doi.org/10.1016/j.ijtst.2023.08.005.

Zhang, Z., Liu, X., 2020. Safety risk analysis of restricted-speed train accidents in the United States. J. Risk Res. 23 (9), 1158–1176.